

Complete SSD Cold Embedding Analysis — All Numbers Measured

Metric	Value
Model	26 embedding tables, D=16, 2,061 MB fp32
Batch size	128 samples
Embedding lookups/batch	53,241 total (51,581 hot + 1,660 cold)
Cold data needed/batch	104 KB (1,660 rows × 64 bytes)
Cold file on SSD	2,037 MB (33,378,031 rows)
Unique 4KB pages/batch	882 (91.7% have only 1 row)
SSD data read/batch	3,528 KB = 3.4 MB (34x amplification)
Wasted bytes	97.1% (read 4KB page for 64B row)
SSD random 4K IOPS (fio)	QD=1: 6,818 (142μs) QD=32: 79,526
Sequential bandwidth	492 MB/s
Overhead: Python/syscall/SSD	0.2ms / 1.8ms / 102ms (98% SSD)
DRAM: embedding lookup	1.46ms (full forward: 4.8ms)
SSD: read + lookup + forward	104ms + 2.4ms + 4.8ms = 112ms
DC: lookup + forward	1.48ms (full forward: 4.4ms) Memory: 10.9 MB (189x)
DC vs DRAM speed	~1.0x (same speed, 3 rounds measured with cache drops)
SSD vs DRAM speed	23x slower